

ME300 - Homework 3 Solutions

1/2

① Given: $L = 58.2 \text{ m} = 1.478 \text{ m}^{+1/2}$ Assume: ideal gas $+2$

a) $D = 9 \text{ in} = 0.2286 \text{ m}^{+1/2}$

$$M = 25 \text{ lb} = 11.34 \text{ kg}^{+1/2}$$

$$P = 2900 \text{ psi} = 19.99 \text{ MPa}^{+1/2}$$

b) ~~$PV = nRT$~~ $m = \frac{M}{N} \rightarrow N = \frac{M}{m} = \frac{11.34}{28.0134} \rightarrow \boxed{N = 0.405 \text{ kmol}}^{+1}$

$$m_{N_2} = 28.0134 \text{ kg/kmol}^{+1}$$

c) $R = \frac{\bar{R}}{m} = \frac{8314.5 \text{ J/kmol}\cdot\text{K}}{28.0134 \text{ kg/kmol}} \rightarrow \boxed{R = 296.8 \text{ J/kg}\cdot\text{K}}^{+1}$

d) $PV = nRT \rightarrow T = \frac{PV}{nR} = \frac{(19.99 \times 10^6)(0.019)}{(11.34)(296.8)} \rightarrow \boxed{T = 112.85 \text{ K}}^{+1}$

$$V = \frac{D^2 L}{4} = 0.019 \text{ m}^3$$

e) $PV = nRT \rightarrow T = \frac{PV}{nR} \rightarrow \text{will be the same temperature b/c all other properties are the same} \rightarrow P, V, N, R$

$$\rightarrow \boxed{T = 112.85 \text{ K}}^{+1}$$

② Given: $P = 19.99 \text{ MPa}$ - absolute pressure

a) $P_{\text{abs}} = P_{\text{gage}} + P_{\text{atm}}^{+3}$

$$P_{\text{gage}} = P_{\text{abs}} - P_{\text{atm}} = 19.99 \times 10^6 - 101325 \text{ Pa}^{+2}$$

$$\boxed{P_{\text{gage}} = 19888675 \text{ Pa}}^{+2}$$

$$b) P_{gaze} = P_{abs} - P_{atm}$$

$$= (19.99 \times 10^6) - 71705.48$$

$$P_{gaze} = 19918294.52 \text{ Pa}$$

$$3) a) R = \frac{R}{M} = \frac{8314.5 \text{ J/kmol-K}}{31.9988 \text{ kg/kmol}} \rightarrow R = 259.84 \text{ J/kg-K}$$

$$b) v = 10 \frac{\text{m}^3}{\text{kg}}$$

$$\bar{v} \rightarrow \bar{v} = \frac{v}{N} \text{ where } v = \frac{V}{M} \text{ and } M = \frac{m}{N}$$

$$v = VM \text{ and } M = mN$$

$$v = v m N$$

$$\rightarrow \bar{v} = \frac{v m N}{N} \rightarrow \bar{v} = v m = (10)(44.01)$$

$$\bar{v} = 440.1 \text{ m}^3/\text{kmol-K}$$

$$c) R = \frac{R}{m} \rightarrow m = \frac{R}{R} = \frac{8314.5 \text{ J/kmol-K}}{188.92 \text{ J/kg-K}} \rightarrow m = 44.01 \frac{\text{kg}}{\text{kmol}}$$

$$d) M = 20 \text{ kg}$$

$$N = \frac{M}{m} = \frac{20 \text{ kg}}{18.01528 \text{ g/kmol}} \rightarrow N = 1.1 \text{ kmol}$$

$$e) \bar{v} = 0.1 \text{ m}^3/\text{mol} \rightarrow v = \frac{\bar{v}}{m} \rightarrow \rho = \frac{m}{\bar{v}} = \frac{28.0134 \text{ g/mol}}{0.1 \text{ m}^3/\text{mol}}$$

$$\rho = \frac{1}{\bar{v}}$$

$$\rho = 280.134 \text{ g/m}^3$$

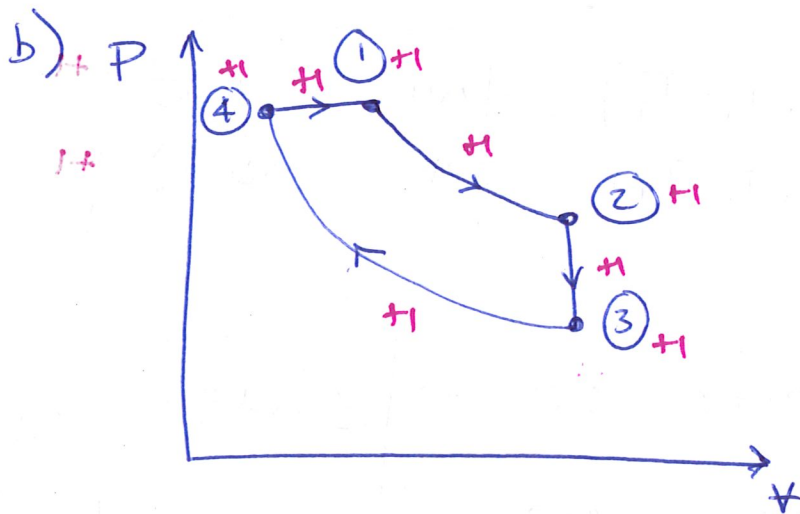
$$\rho = 0.28 \text{ kg/m}^3$$

④ Ideal gas $\rightarrow$ molecules are very far apart $v_{mol} \ll v_{bw mol}^{2/2}$
 $\rightarrow$ all collisions are elastic

$$\Rightarrow PV = NRT$$

⑤ a)

Process	T	P	V	
1-2	const.	$\downarrow$	$\uparrow$	$\Rightarrow PV = MRT$ +1 each
2-3	$\downarrow$	$\downarrow$	const.	
3-4	const.	$\uparrow$	$\downarrow$	
4-1	$\uparrow$	const.	$\uparrow$	



⑥ Given: $C_v = 700 \text{ J/kg-K}$ $T_1 = 300 \text{ K}$ Assume: ideal gas $+1$
 $C_p = 900 \text{ J/kg-K}$ $T_3 = 800 \text{ K}$ $C_p = \text{const.}$ $+1$

Process 1-2: $\Delta u = C_v \Delta T$ $+3$ $\rightarrow$ $\Delta u = 0 \text{ J/kg}$ $+1$
 $\Delta h = C_p \Delta T$ $+3$ $\rightarrow$ $\Delta h = 0 \text{ J/kg}$ $+1$

Process 2-3: $u_3 - u_2 = C_v (T_3 - T_2) = 700(800 - 300) \Rightarrow \Delta u = 350 \frac{\text{kJ}}{\text{kg}}$ $+1$
 $h_3 - h_2 = C_p (T_3 - T_2) = 900(800 - 300) \Rightarrow \Delta h = 450 \frac{\text{kJ}}{\text{kg}}$ $+1$

Process 3-4: $\Delta u = C_v \Delta T$ $+1$ $\rightarrow$ $\Delta u = 0 \text{ J/kg}$ $+1$
 $\Delta h = C_p \Delta T$ $+1$ $\rightarrow$ $\Delta h = 0 \text{ J/kg}$ $+1$

Process 4-1: $u_1 - u_4 = C_v (T_1 - T_4) \rightarrow \Delta u = -350 \frac{\text{kJ}}{\text{kg}}$ $+1$
 $h_1 - h_4 = C_p (T_1 - T_4) \rightarrow \Delta h = -450 \frac{\text{kJ}}{\text{kg}}$ $+1$

$\Rightarrow \sum \Delta u = 0 + 350 + 0 - 350 \rightarrow \sum \Delta u = 0 \text{ J/kg}$ $+1$
 $\sum \Delta h = 0 + 450 + 0 - 450 \rightarrow \sum \Delta h = 0 \text{ J/kg}$ $+1$

$\Rightarrow$ This makes sense b/c this is a cycle and the system must return to its original state, so all changes in properties like u and h must be zero. $+2$